Parkinson’s Disease: A Progressive Brain Disorder Beyond Movement

Parkinson’s disease (PD) is a neurological disorder that progresses over time and primarily affects movement. However, Parkinson’s disease is not merely a movement disorder; it can also affect mental health, sleep, cognitive functions, pain, and various other bodily functions. As the disease progresses, it can significantly affect a person’s daily life, independence, and quality of life. Worldwide, disability and mortality associated with Parkinson’s disease are increasing. According to estimates from 2019, more than 8.5 million people worldwide were living with Parkinson’s disease.

The underlying basis of Parkinson’s disease involves damage to and gradual loss of certain nerve cells in the brain. In particular, dopamine-producing nerve cells in a brain region called the *substantia nigra* play an important role. Dopamine is an important chemical messenger involved in the signals that enable the brain to produce smooth and purposeful movements. By the time Parkinson’s disease symptoms appear, a substantial proportion of the dopamine-producing cells in the *substantia nigra* have already been lost. In addition, most people with Parkinson’s disease have Lewy bodies, which are abnormal accumulations of a protein called alpha-synuclein, in damaged nerve cells. However, the exact role of these structures in the development of the disease has not yet been fully explained.

The most well-known symptoms of the disease are related to movement. Tremor, muscle rigidity, slowed movement (bradykinesia), and postural and balance disturbances are among the main motor symptoms of Parkinson’s disease. Tremor often begins in the hand and may be particularly noticeable when the person is at rest. Muscle rigidity can make movement difficult and may cause pain. Bradykinesia, on the other hand, can cause a person to perform everyday activities that were previously easy, such as washing, dressing, or walking, more slowly. As the disease progresses, walking problems such as taking small steps, leaning forward, reduced natural arm swing, and brief episodes of “freezing” during movement may also occur.

However, the effects of Parkinson’s disease are not limited to the motor system. Non-motor symptoms such as sleep problems, cognitive impairment, mental health problems, pain, fatigue, changes in speech, difficulty swallowing and chewing, and urinary and digestive problems may also occur. Some individuals may develop dementia in the later stages of the disease. Therefore, Parkinson’s disease should not be regarded as a condition that manifests only through tremor.

The exact cause of Parkinson’s disease is unknown in most individuals. Nevertheless, age is an important risk factor, and the disease occurs more commonly in older adults. It affects men more frequently than women. A family history and certain genetic characteristics may also increase the risk. In addition, there is evidence that environmental exposures such as pesticides, certain industrial solvents, and air pollution may be associated with an increased risk of Parkinson’s disease. Genetic susceptibility and environmental factors are thought to play a role together.

There is no single test that can definitively establish the diagnosis of Parkinson’s disease. Diagnosis is primarily based on the person’s medical history and neurological examination. Certain laboratory and imaging methods may help evaluate other disorders that can cause similar symptoms or support the diagnosis. Therefore, evaluating the patient’s symptoms together with neurological findings is important in the diagnosis of Parkinson’s disease.

There is currently no definitive treatment that completely eliminates Parkinson’s disease. However, medications, surgical approaches, and rehabilitation interventions can help control symptoms. Levodopa/carbidopa is one of the most effective and commonly used treatments for improving motor symptoms. In some patients, surgical approaches such as deep brain stimulation may also be used. In addition, physiotherapy, gait and balance training, strength exercises, and other rehabilitation interventions can support a person's functioning and quality of life.

In conclusion, Parkinson's disease is not merely a tremor of the hands or a slowing of movement. It is a progressive and complex neurological disorder associated with the loss of dopamine-producing cells in the brain and characterized by a wide range of both motor and non-motor features. The goal of treatment is not only to improve movement but also to preserve the person's daily functioning, independence, and quality of life as much as possible.

References:

1) World Health Organization. (2023, August 9). Parkinson disease. https://www.who.int/news-room/fact-sheets/detail/parkinson-disease

2) World Health Organization. (2022, June 14). Parkinson disease: A public health approach: Technical brief. https://www.who.int/publications/i/item/9789240050983

3) National Institute of Neurological Disorders and Stroke. (n.d.). Parkinson's disease. National Institutes of Health. https://www.ninds.nih.gov/health-information/disorders/parkinsons-disease

Ali Bocakkıtay/ Founder, Wionpav Labs/ Medical Student, Faculty of Medicine, Selçuk University